



SYMBIOSIS

SCHOOL OF ECONOMICS

DETERMINANTS OF FEMALE LABOUR FORCE PARTICIPATION:

A Cross Country Analysis of Non High Income Economies (2022)

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Submitted by

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1. Abstract

The research investigates factors that affect female labour force participation (FLFP) in 80 non-high-income countries through cross-sectional data from 2022 which The World Bank provided through its World Development Indicators (WDI) database. The research aims to evaluate how demographic characteristics and economic conditions together with regional factors affect FLFP in developing and middle-income countries. The research uses Ordinary Least Squares (OLS) regression to assess how urban population percentage and log-transformed fertility rate and log-transformed female life expectancy and South Asia regional dummy variables affect study results.

The findings reveal that all independent variables show statistical significance. The feminisation U-hypothesis shows that urbanisation decreases FLFP according to the study results. The fertility rate shows a strong negative relationship because increasing caregiving responsibilities prevent women from showing their participation rate in the labor market. Female life expectancy shows a negative relationship because demographic and structural factors affect the results. The South Asia dummy demonstrates that South Asian countries face institutional and socio-cultural barriers which lead to their citizens having lower participation rates.

The model demonstrates a moderate fit, and diagnostic tests confirm its robustness. The study results show that demographic and structural factors determine female labour force participation. The research findings enable policymakers to develop specific programs which will advance gender equality while promoting economic growth that benefits all people in developing nations.

2. Introduction

2.1 Background and Motivation

The world faces a critical issue concerning gender inequality in the workforce which has emerged as an international problem that affects developing nations and middle-income countries. Women face major obstacles that prevent them from joining and staying in the workforce despite improvements in education and healthcare and human development indicators.

Economic growth combined with urbanization should lead to higher female workforce participation according to traditional development theory. Urban areas in developing nations show a pattern where women stop working in informal agricultural jobs to enter cities which present stronger challenges for them to find formal work. The U-shaped feminization hypothesis describes this phenomenon because it shows that female labor force participation decreases first before rising again throughout economic advancement.

The phenomenon develops through multiple elements which include urbanization and fertility rates together with female life expectancy and the varying institutional frameworks of different regions. The higher fertility rates lead to increased caregiving obligations which prevent women from pursuing employment opportunities. The process of urbanization creates immediate employment restrictions because existing labor market systems face operational limitations. Female life expectancy represents health status together with life duration, which influences work availability decisions, while regional characteristics particularly found in South Asia show the persistent cultural and institutional obstacles that impede women's workforce participation.

Policymakers and development organizations together with economists need to comprehend these relationships. The improved FLFP leads to better economic productivity because it makes use of existing workforce resources, and it helps to decrease poverty while enhancing family living standards and empowering women.

2.2 Theoretical Framework

Theoretical foundations of this research study depend on three established economic frameworks. Becker's (1965) time allocation model establishes the foundation for research which shows how demographic constraints through fertility limits women's work capacity by increasing their market work opportunity costs compared to home production. Goldin's (1995) feminisation U-curve describes how economic growth and urban development create an at-first decrease in women's workforce participation which occurs because women transition from agricultural jobs to formal employment where they face greater entry obstacles. The human capital model of health developed by Grossman (1972) demonstrates that female life expectancy serves as a direct measure for determining women's productivity and career span thus making it a necessary health capital metric which should be used together with standard economic indicators. The empirical model tested in this study uses three frameworks as its theoretical foundation because every separate variable connects to one or more of these three theories.

2.3 Research Objectives

The research investigates the factors that influence female employment in 80 countries that do not qualify as high-income nations by analyzing cross-sectional information from 2022. Researchers examine non-high-income nations to create a study group that maintains similar characteristics while displaying how developing countries women join the workforce. The study uses Ordinary Least Squares (OLS) regression to determine how urban population share and log-transformed fertility rate and log-transformed female life expectancy and the South Asia regional dummy variable affect FLFP.

3. Literature Review

The research on female labor force participation (FLFP) extends across multiple fields which include development economics, labor economics, demographic research, and institutional economics. This section reviews studies which directly connect to the variables used in this research.

3.1 Urbanisation and FLFP

Goldin (1995) introduces the feminisation U-curve which shows that FLFP declines during early industrialisation as women transition from agricultural work to urban areas where they encounter greater barriers to formal employment until advanced economies reach their peak. Mammen and Paxson (2000) provide strong cross-country empirical support because they show that middle-income urbanising economies always display the lowest female participation rates. The research conducted by Mitra and Tripathi (2024) demonstrates a significant negative impact of urbanisation on female-to-male labour force participation throughout developing economies using World Bank data obtained from 217 countries between 1991 and 2022. The analysis conducted by the researchers examines a mixed-income global sample but this study provides new insights through its estimation of urbanisation effects within developing countries that experience the most active U-curve effect.

3.2 Fertility Rate and FLFP

Becker (1965) establishes the theoretical foundation through his time allocation model which shows that every additional child born to a woman decreases her ability to work. The first child imposes a greater restriction on women than the fifth child according to the model which needs log transformation to study fertility in this research. Angrist and Evans (1998) provide evidence which establishes that women decrease their work hours when they have more children. Bloom, Canning, Fink, and Finlay (2009) used cross-country panel data to show that abortion legislation created a strong negative causal relationship which showed that each birth leads to a two-year decrease in a woman's labor supply. The researchers established cross-country causal evidence to support their decision to use $\log(\text{Fertility Rate})$ as a fundamental factor in their current study.

3.3 Female Life Expectancy and FLFP

Grossman (1972) describes health as human capital that affects the amount of work done by individuals. Canning and Schultz (2012) show that when women experience better health their chances of joining the workforce rise while Kaur and Sharma (2023) present evidence from India that supports this claim. The research conducted by Goldin (1995) and Mammen and Paxson (2000) demonstrates that increasing life expectancy in developing economies leads to decreased FLFP because income effects take precedence.

The majority of cross-country research studies human capital through female education which serves as the main indicator while health degrees of human capital remain unmeasured (Thévenon, 2013; Klasen, 2019). The two human capital elements education and health require health factors to be measured because they display distinct human capital elements.

Becker (1965) establishes the theoretical foundation through the time allocation model which demonstrates how household duties compete against available work hours. The model implies diminishing marginal effects of additional children, which makes using log-transformed fertility data necessary. Angrist and Evans (1998) provide empirical evidence which shows that additional children decrease female labor supply. Bloom et al. (2009) present cross-country causal proof which demonstrates a strong negative connection between fertility rates and female workforce participation.

According to Grossman (1972) people consider health as human capital which affects their ability to work. Female life expectancy serves as a comprehensive measure of women's health across the lifecycle and is more informative than proxies such as maternal or infant mortality rates. This motivates the inclusion of log(female life expectancy) as an independent variable in the study.

3.4 Regional Institutional Factors: South Asia Dummy

Klasen and Pieters (2015) identify what they call the India paradox because between 2000 and 2010 female labor force participation decreased while Indian GDP experienced fast growth and female education increased together with fertility rates declining. The researchers observe that South Asian households combine increased income with patriarchal social systems to demonstrate status by keeping their women at home.

All current cross-country FLFP research studies either disregard South Asian nations or treat them as identical to other developing nations or they use country fixed effects to handle regional impacts which do not measure anything. No study using cross-country OLS for South Asian FLFP penalty estimation exists which controls urbanization and fertility and health factors in its examination of developing countries. The present study uses South Asia dummy variable because it defines institutional deficit at 18.7 percentage points which standard economic variables fail to establish.

3.5 Research Gap

This study identifies three gaps in current literature which require investigation. The cross-country FLFP models currently used in research ignore health as an independent factor because they only use education as a human capital measurement which fails to recognize that health and education represent two distinct aspects of women's labor market abilities (Grossman, 1972; Canning and Schultz, 2012). The negative relationship between urbanization and FLFP in developing countries has been established through research (Mitra and Tripathi, 2024) yet researchers have not analyzed its effects through a non-high-income sample which restricts income variation while solving heteroskedasticity issues. No existing parsimonious specification simultaneously quantifies the South Asian FLFP institutional deficit while controlling for economic, demographic, and health determinants (Klasen and Pieters, 2015).

The research uses a four-variable OLS model to study three research gaps across 80 non-high-income countries. The log transformations provide solutions to both non-linear

problems and heteroskedasticity issues. The sample restriction based on theoretical grounds creates a unified sample of developing countries. The South Asia dummy explicitly quantifies the institutional FLFP penalty previously absorbed into fixed effects. The combined research creates an evaluation of FLFP determinants which exceeds existing cross-country models through its theoretical foundation and econometric analysis.

4. Data and Variables

4.1 Data Source and Sample

The study relies on the 2022 World Development Indicators (WDI) dataset provided by the World Bank. This dataset pools official statistics from notable organizations like the ILO, United Nations, and national statistical bodies, ensuring cross-country comparability. For this analysis, the sample includes 80 nations classified by the World Bank as having a GNI per capita below US\$13,845 for 2022. High-income countries are intentionally left out, as their Female Labour Force Participation Rate (FLFP) tends to exhibit different patterns, and combining them with lower-income countries might lead to statistical inconsistencies, such as heteroskedasticity.

4.2 Variable Definitions

Dependent Variable

Female Labour Force Participation Rate (FLFP): Percentage of women aged 15 and above actively participating in the workforce. (WDI: SL.TLF.CACT.FE.ZS).

Independent Variables

- a. **Urban Population (%):** Represents population living in urban areas (WDI: SP.URB.TOTL.IN.ZS). The expected sign here is negative, aligning with the feminisation U-hypothesis, which suggests urbanization initially decreases female workforce participation before reversing at advanced stages of development. (Mitra and Tripathi, 2024).
- b. **log(Fertility Rate):** Natural logarithm of total fertility rate (WDI: SP.DYN.TFRT.IN). The expected sign is negative, showing how caregiving responsibilities potentially lower workforce participation (Becker, 1965; Bloom et al., 2009).
- c. **log(Female Life Expectancy):** Natural logarithm of female life expectancy at birth (WDI: SP.DYN.LE00.FE.IN). The expected sign is negative for developing economies, a longer life expectancy may correlate negatively with FLFP due to intertwined social and economic factors. (Goldin, 1995; Grossman, 1972; Canning and Schultz, 2012).

- d. South Asia Dummy:** Binary variable (1 = South Asia: Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan, Sri Lanka; 0 = others). The expected sign is negative, women in this region often face unique societal barriers to employment (Klasen and Pieters, 2015).

Table 1: Variable Summary

Variable	Description	WDI Code	Role
FLFP (%)	Female LFP rate, ages 15+	SL.TLF.CACT.FE.ZS	Dependent
Urban Pop (%)	Urban population share	SP.URB.TOTL.IN.ZS	Independent
log(Fertility Rate)	log of total fertility rate	SP.DYN.TFRT.IN → log	Independent
log(FemLifeExp)	log of female life expectancy	SP.DYN.LE00.FE.IN → log	Independent
South Asia Dummy	South Asia regional dummy (0/1)	World Bank classification	Dummy

Note: All continuous variables sourced from World Bank WDI, 2022. South Asia dummy based on World Bank regional classification (World Bank, 2023).

5. Research Methodology

5.1 Estimation Strategy

The study uses cross-sectional Ordinary Least Squares (OLS) regression as its main research method to examine the determinants of female labour force participation (FLFP) across non-high-income countries. OLS is used to estimate the relationship between FLFP and its independent variables by fitting a linear model to the data.

To ensure the reliability of the results, diagnostic testing is conducted. Heteroskedasticity is tested using the White test, which confirms that the error terms have constant variance, supporting the validity of the regression estimates.

5.2 Model Specification

The regression model is:

$$FLFP_i = \beta_0 + \beta_1(Urban_i) + \beta_2(\log Fertility_i) + \beta_3(\log Female Life Expectancy_i) + \beta_4(South Asia_i) + \varepsilon_i$$

Where:

- β_0 = intercept
- β_1 – β_4 = coefficients of the independent variables
- ε_i = error term

5.3 Log Transformation Rationale

The research uses log-transformation on two variables which are fertility rate and female life expectancy because three reasons necessitate this transformation. The first requirement needs both variables to show their theoretical non-linear relationship because they create diminishing marginal effects on FLFP. The second requirement needs extreme values to be reduced, which helps to solve the data skewness issue. The third requirement establishes that log specifications will boost econometric results because they provide variance stabilization and model dependability enhancement.

5.4 Model Selection Process

The final model is obtained through a systematic testing process. Initial estimations using the full sample showed heteroskedasticity, likely due to income heterogeneity. Restricting the sample to non-high-income countries improved model consistency.

Subsequently, alternative model specifications were tested, and the final model was selected based on three criteria: absence of heteroskedasticity, statistical significance of all variables ($p \leq 0.05$), and inclusion of a relevant regional dummy. This specification was found to provide the most robust and reliable estimates.

Table 2: Model Selection Process

Stage	Variables	Sample	R ²	Heteroskedasticity	Significance of Variables	Status
1	Fertility, GDP Growth, Education, Urban, Unemployment	N=132	0.08	Yes	No	Rejected
2	log(GDPpc), FemLifeExp, MENA, WomenParl, Urban	N=132	0.38	Yes	No	Rejected
3	All combinations tested (21+ models)	N=132	Various	Yes	Various	Rejected
4	FemLifeExp + log(GDPpc) + WomenParl (no dummy)	N=80	0.244	No	Yes	Rejected (no dummy)
5	Urban + log(Fert) + log(FLE) + SouthAsia	N=80	0.322	No	Yes	Final Model

Note: Final model selected on the basis of highest R², all variables significant at $p < 0.05$, no heteroskedasticity (White $p = 0.121$), and all VIFs < 3 .

Source: Excel workbook

5.5 Hypothesis Testing

The statistical significance and validity of the model are evaluated using the following tests:

Hypothesis Test for slope coefficients :

- $H_0: \beta_i = 0$ (slope coefficient is 0 ; independent variable does not affect the dependent)

- $H_1: \beta_i \neq 0$ (slope coefficient is not 0 ; independent variable affects the dependent)
- Decision rule: Reject H_0 if p-value < 0.05 .

Hypothesis Test for Overall Significance (F test) :

- H_0 : The overall model is not significant; independent variables do not explain the variation in the dependent variable. ($\beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$)
- H_1 : The overall model is significant. At least one independent variable explains the variation in the dependent variable.
- Decision rule:
 - $F_{cal} > F_{critical}$ (Reject H_0 the model is significant)
 - $F_{cal} < F_{critical}$ (Fail to reject H_0 ; the model is insignificant)

Hypothesis Test for Heteroskedasticity (White's tests):

- H_0 : Residuals have constant variance (homoscedasticity); heteroscedasticity is not present.
- H_1 : Residuals do not have constant variance (heteroscedasticity is present).
- Decision rule: Reject H_0 if p-value < 0.05 .

Hypothesis Test for Multicollinearity (Variance Inflation Factor - VIF):

- H_0 : No multicollinearity is present among independent variables.
- H_1 : Multicollinearity is present among independent variables.
- Decision Rule :
 - $VIF < 5$: No multicollinearity concern
 - $VIF 5-10$: Moderate concern
 - $VIF > 10$: Serious multicollinearity problem

6. Results and Findings

6.1 Descriptive Statistics

Table 3 : Descriptive Statistics

Variable	N	Mean	Std Dev	Min	Max	Median	Role
FLFP (%)	80	48.292	14.754	13.020	77.173	51.177	Dependent
Urban Population (%)	80	56.244	19.065	16.715	92.405	57.926	Independent
Fertility Rate (raw)	80	2.453	0.935	1.221	4.634	2.203	Reference only
log(Fertility Rate)	80	0.831	0.363	0.200	1.533	0.790	Independent
Female Life Exp (raw)	80	74.522	5.073	61.750	82.546	76.170	Reference only
log(Female Life Exp)	80	4.309	0.070	4.123	4.413	4.333	Independent
South Asia Dummy	80	0.088	0.284	0.000	1.000	0.000	Independent / Dummy

The study uses descriptive statistics to present an overview of the main traits that define its research variables. The Female Labour Force Participation (FLFP) mean value for the 80-country sample stands at 48.29% which displays substantial differences in female workforce participation across different countries. The FLFP range extends from 13.02% to 77.17% which demonstrates significant differences between countries regarding their female workforce participation rates.

The average urban population share across all countries reaches 56.24% while actual urban population shares vary from 16.71% to 92.41% which demonstrates different levels of urban development between nations. The average fertility rate for the group stands at 2.45 births per woman based on the log mean value of 0.831 which demonstrates different demographic patterns throughout the studied population.

The average female life expectancy reaches 74.52 years while the standard deviation of 5.07 years shows that health outcomes among females remain more consistent throughout the studied population. The log-transformed value has a mean of 4.309.

Standard deviations provide a further demonstration of how these variables display different levels of variability. Urban population demonstrates high dispersion through its value of 19.07 whereas log fertility and log female life expectancy show moderate and low variation through their respective values of 0.363 and 0.070.

The South Asia dummy variable has a mean value of 0.088 which shows that about 8.8% of the sample population (7 out of 80 countries) belongs to South Asia region. The South Asia dummy variable shows regional differences in female labor force participation through its classification of South Asian countries.

The diverse FLFP patterns which exist between nations together with their unique FLFP determinants create a suitable environment for econometric analysis that investigates how each factor affects FLFP independently.

6.2 Regression Analysis

Table 4 : OLS Regression Results

Variable	Coefficient (β)	Std Error	t-statistic	p-value	95% CI Lower	95% CI Upper
Intercept	642.5595	142.0926	4.5221	0.0000	359.4965	925.6224
Urban Population (%)	-0.1780	0.0892	-1.9962	0.0495	-0.3556	-0.0004
log(Fertility Rate)	-25.7210	6.0572	-4.2463	0.0001	-37.7876	-13.6543
log(Female Life Exp)	-130.2610	32.4673	-4.0121	0.0001	-194.9392	-65.5829
South Asia Dummy	-18.6750	5.3100	-3.5170	0.0007	-29.2530	-8.0970

The study uses regression analysis to investigate how various factors determine female labor force participation (FLFP) in the study population. The R^2 value of 0.3223 indicates that

approximately 32.23% of the variation in FLFP is explained by the independent variables, while the Adjusted R^2 of 0.2861 confirms a reliable model fit after accounting for the number of predictors. The values show moderate strength which arises as common results across different countries because hidden cultural and institutional elements affect their results.

The intercept value of 642.56 serves as a basic standard measurement. The coefficient for urban population is -0.178 ($p = 0.0495$), which establishes that every one percentage point rise in urban population share results in a 0.178 percentage point decline of FLFP according to the feminisation U-hypothesis.

The coefficient for $\log(\text{fertility rate})$ is -25.721 ($p < 0.001$), which shows a percentage increase in fertility is associated with a decline in FLFP, reflecting increased caregiving responsibilities.

The $\log(\text{female life expectancy})$ coefficient shows a value of -130.261 ($p < 0.001$), which demonstrates that higher female life expectancy leads to decreased FLFP according to the wealth-health-withdrawal effect observed in developing nations.

The South Asia dummy (-18.675 , $p = 0.0007$) shows that FLFP is about 18.7 percentage points lower in South Asian countries compared to others, highlighting strong regional institutional constraints.

6.3 Significance of Slope Coefficients

Table 5 : Significance of Slope Coefficients

Model Fit Statistics		Explanation
R²	0.3223	32.2% of variation in FLFP explained by the model
Adjusted R²	0.2861	Penalised for number of regressors — confirms no overfitting
F-statistic	8.916	Model is jointly significant
F p-value	0.00000605	$p < 0.05$ — reject H_0 of overall insignificance
Observations (N)	80	Non-high-income countries, cross-section 2022
Degrees of freedom (residual)	75	$N - k - 1 = 80 - 4 - 1$

The research findings demonstrate statistical significance which proves that at least one studied independent variable affects female labour force participation (FLFP) outcomes.

P-value (F-test) = 0.00000605

$\therefore P < 0.05$

$\therefore$ The null hypothesis of overall insignificance is rejected.

The researchers evaluated slope coefficients through t-tests which tested urban population log(fertility rate) log(female life expectancy) and south asia dummy variables at a 0.05 significance level. The p-values for all variables are below 0.05—Urban (0.0495), log(Fertility) (0.0001), log(Female Life Expectancy) (0.0001), and South Asia (0.0007).

The null hypothesis stating that all coefficients equal to zero has been rejected for every variable. The study results show that every independent variable statistically impacts FLFP. The study results demonstrate that urbanization and fertility and health conditions and regional factors determine female labor force participation.

6.4 Significance of Overall Model

The research findings demonstrate statistical significance which proves that at least one studied independent variable affects female labour force participation (FLFP) outcomes.

F-calculated = 8.916 and Significance F (p-value) = 0.00000605

$\therefore P < 0.05$

$\therefore$ The null hypothesis of model insignificance is rejected.

The ANOVA test results demonstrate that the regression model shows extremely high significance. The F-statistic value of 8.916, with a p-value far below 0.05, indicates that the independent variables jointly explain a significant portion of the variation in FLFP.

The results show that the model achieves better performance than a model which uses no predictor variables. All independent variables demonstrate p-values under 0.05 which proves that each one plays an essential role in the explanation of female labour force participation.

6.5 Test for Heteroscedasticity

Table 6 : Results for White's Test

Results for White's Test	
Auxiliary R ²	0.238432
N (observations)	80
k (auxiliary regressors)	13
LM statistic ($N \times R^2$)	19.0746
Chi-square p-value	0.1208
F-statistic	1.5895
F p-value	0.1108

Residuals in the model show constant variance, meaning there's no evidence of heteroscedasticity.

White's test was used to check for this issue.

The calculated test statistic came out as $LM = 19.0746$ ($N \times R^2 = 80 \times 0.2384$), and the p-value was found to be 0.1208. Since the p-value exceeds 0.05, the null hypothesis of homoskedasticity holds.

There's no sign of heteroscedasticity in this dataset. The findings back up the claim of uniform variance across residuals. Because of this, the OLS estimates are considered dependable, and no adjustments are necessary.

6.6 Test for Multicollinearity

Table 7 : Multicollinearity Test

Variable	VIF	Interpretation
Urban Population (%)	1.469	VIF=1.47 — minimal multicollinearity
log(Fertility Rate)	2.457	VIF=2.46 — mild, well within acceptable range
log(Female Life Exp)	2.656	VIF=2.66 — mild, well within acceptable range
South Asia Dummy	1.159	VIF=1.16 — minimal, dummy virtually uncorrelated

All VIF values are well below the threshold of 5, indicating no evidence of multicollinearity. Despite the moderate correlation between log(Fertility) and log(Female Life Expectancy) ($r = -0.766$), the VIF values confirm that this does not pose a serious multicollinearity issue.

Therefore, the coefficient estimates are reliable and not inflated due to collinearity. Each variable contributes unique information to the model, and no corrective measures are required.

6.7 Theoretical Validation

Table 8 : Theory Summary

Variable	Theory	Expected Sign	Actual Sign	β Value	p-value	Theory Confirmation
Urban Population	Goldin (1995) — U-curve	—	—	-0.178	0.0495	Yes
log(Fertility Rate)	Becker (1965) — Time allocation	—	—	-25.721	<0.001	Yes
log(Female Life Exp)	Grossman (1972) — conditional	—	—	-130.26	<0.001	Yes
South Asia Dummy	Klasen & Pieters (2015) — institutional	—	—	-18.675	0.0007	Yes

The regression findings lend strong empirical support to all three theoretical frameworks on which this study is based. The feminisation U-curve by Goldin (1995) is confirmed by the negative coefficient on urbanisation, thus validating the position that developing economies continue to be on the downward slope of the U-curve in 2022. The time allocation model by

Becker (1965) is confirmed by the strong negative fertility effect, the largest effect among all the variables, thus validating demographic constraints as the primary driver for FLFP. The health capital model by Grossman (1972) is conditionally confirmed by the significance and economic relevance of the coefficient on female life expectancy, although its negative sign is consistent with the context-specific findings for developing economies by Goldin (1995). The significance of the dummy on the South Asian economies validates the theoretical prediction by Klasen and Pieters (2015) on the structural FLFP deficit in the region, not captured by the continuous variables. All four theoretical predictions are thus empirically validated by the findings on the developing economy sample.

7. Conclusion

7.1 Summary of Findings

Researchers investigated the reasons behind female labor force participation in 80 different countries that do not belong to high-income nations using cross-sectional OLS regression for the year 2022. All four independent variables which include urban population and fertility rate and female life expectancy and the South Asia dummy show statistically significant effects on FLFP results. Urbanization and fertility and female life expectancy create negative effects on FLFP whereas the South Asia dummy shows that participation rates in that area exist at much lower levels than other regions. The most powerful demographic factor which controls female labor force availability between these elements shows that fertility rate represents the most effective demographic element.

The study results demonstrate that female labor force participation decreases because of increased caregiving duties and urban labor market restrictions and strong socio-cultural obstacles. South Asian countries demonstrate extremely low female labor force participation because of institutional and cultural barriers which exist beyond economic explanations. The demographic and structural and institutional factors present in different countries create variations in female labor force participation rates according to the research findings.

7.2 Policy Implications

The study results create vital policy implications which support the development of gender-inclusive economic growth according to the study results. The government should develop programs which improve childcare access and create job opportunities for women while they remove institutional constraints especially in South Asia. The policies which govern family planning and labor market access will serve as critical drivers that boost female workforce participation rates.

7.3 Limitations of this Research

The study has specific weaknesses which affect its outcomes. Researchers can establish causal relationships only through cross-sectional data which covers one year and the study used a narrow selection of variables because it omitted educational attainment and income levels and social norms. The findings from this research face limitations in their general applicability because of these research constraints.

Upcoming studies can amplify this inquiry by using panel data and adding educational data and institutional quality information and performing studies that focus on specific regions. Such approaches would provide a deeper and more comprehensive understanding of the determinants of female labour force participation.

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